Maths for Computing — Phase 5: Mock Solved and Harder Practice Exam

Nov 10, 2026 · @Hirushan Perera

How to use this phase and exam-day strategy

This phase brings everything together. It has three parts:

1. **Every mock question you sent, solved**, with the phase each topic is taught in. (Question 16 wasn't in your screenshots, so it isn't covered.)
2. **A one-page method sheet** to read the morning of the exam.
3. **A harder practice exam** of 26 mixed questions with full answers. Try it under timed conditions: about 60 minutes.

Mock answer key at a glance

Q	Answer	Q	Answer	Q	Answer
1	A	9	B	18	A
2	D	10	B	19	C (109)
3	B (4)	11	C	20	B
4	D	12	D	21	A
5	B (16)	13	A	22	A
6	B (7th)	14	D (26)	23	C (nfz)
7	C	15	C (Row 3)		
8	B (∅)	17	B		

Exam-day strategy

- **First 2 minutes:** on scrap paper, write the four logic tables, the powers of 2 up to 1024, and the alphabet numbered a = 0 to z = 25. Then you never have to recall them under pressure.
- **Do the quick ones first.** Binary, mod, row counting, function or not, set builder and interval questions take 1–2 minutes each. Leave long truth tables and recursion for a second pass.

- **Points are not equal.** Your mock had questions worth 2 to 8 points. Don't spend 10 minutes on a 2-point question.
- **Use the options.** In multiple choice you can often eliminate. Check the first letter of a cipher, the parity (odd/even) of a binary number, the slope of a line, or plug a test value into an inverse.
- **Write every step** of recursion and truth tables. Most lost marks come from skipped steps, not hard ideas.
- **Read the brackets.** { } versus [ ] versus ( ) changes the answer. So does ∈ versus ⊆, and "in the last block" versus "left in the last block".
- **Never leave a question blank.** If you're stuck, eliminate what you can and guess.

Mock questions 1 to 8

Q1 (4 pts) — Venn diagram. Answer: A — Phase 2 §5

From the diagram: A = {a, b, c, e, f}, B = {d, e, f}. So A \ B = {a, b, c} and B \ A = {d}.

- (A) (A \ B) ∩ (B \ A) = {a, b, c} ∩ {d} = { }. **True.**
- (B) (B \ A) ∪ {a, b, c} = {a, b, c, d} ≠ B.
- (C) A \ (A ∩ B) = {a, b, c} ≠ {e, f}.
- (D) (A ∪ B) ∪ {d} = {a, b, c, d, e, f} ≠ A.

Q2 (6 pts) — Perpendicular line. Answer: D — Phase 3 §6

1. Slope of BC: (−4 − 5)/(−3 − 2) = −9/−5 = 9/5.
2. Perpendicular slope: −5/9.
3. Through A(4, −3): −3 = (−5/9)(4) + b → b = −3 + 20/9 = −7/9.
4. **y = −(5/9)x − 7/9.**

Q3 (4 pts) — Count FALSE in (p ∨ ¬p) ⇒ (q ∧ ¬q). Answer: B (4) — Phase 4 §5

The left is always T and the right is always F. T ⇒ F = F in all 4 rows.

Q4 (2 pts) — When is R a function? Answer: D — Phase 3 §3

R = {⟨a, 0⟩, ⟨b, 1⟩, ⟨c, 1⟩}. It needs domain exactly {a, b, c} (each used once) and a co-domain containing both 0 and 1. Only D: A = {a, b, c}, B = {1, 0}. Option A fails because d has no output; B and C fail because 1 isn't in B.

Q5 (3 pts) — Rows in the truth table. Answer: B (16) — Phase 4 §4

Different letters: p, q, r, s, so n = 4 and $2^4 = 16$.

Q6 (4 pts) — IP address wrap-around. Answer: B (7th) — Phase 1 §5

147 + 10 = 157. The pool has 150 addresses, so 157 mod 150 = 7. Counting by hand: 148, 149, 150, then 1 to 7. That's 10 steps.

Q7 (5 pts) — When is $\neg(p \Rightarrow q)$ true? Answer: C — Phase 4 §2

$p \Rightarrow q$ is false only for p = T, q = F. So its negation is true only then.

Q8 (5 pts) — $(1, 10) \cap \{10\}$. Answer: B ($\varnothing$) — Phase 2 §8

The round bracket means 10 is not in (1, 10). The only candidate, 10, is excluded, so the intersection is empty.

Mock questions 9 to 15

Q9 (3 pts) — Which subset statement is true? Answer: B — Phase 2 §8

- (A) $[1, 2] \subseteq \{1, 3\}$: 1.5 is in the interval but not in the set. False.
- (B) $[1, 2] \subseteq [1, 3)$: every number from 1 to 2 lies in [1, 3). **True.**
- (C) $[1, 2] \subseteq (1, 3)$: 1 is excluded from (1, 3). False.
- (D) $\{1, 3\} \subseteq [1, 2]$: 3 is not in [1, 2]. False.

Q10 (5 pts) — $\{x \in \mathbb{Z} : x^3 = -8\}$. Answer: B ($\{-2\}$) — Phase 2 §4

$(-2)^3 = -8$, and no other number cubes to −8 ($2^3 = +8$). −2 is an integer.

Q11 (4 pts) — Set builder for $(1, 100]$. Answer: C — Phase 2 §8

"(" means $1 < x$; "]" means $x \leqslant 100$.

Q12 (4 pts) — Function with domain {p, q}, co-domain {0, 1}. Answer: D — Phase 3 §3

$\{\langle p, 1\rangle, \langle q, 0\rangle\}$: each input once. B and C give one input two outputs; A pairs the wrong sets.

Q13 (6 pts) — $A = [4, 25)$; $\{x \in A : x^2 \in A\}$. Answer: A ($[4, 5)$) — Phase 2 §8

$4 \leqslant x^2 < 25$ gives $2 \leqslant x < 5$. Combined with $x \geqslant 4$: $[4, 5)$. The 5 is excluded because $5^2 = 25$ is not in A. Option D, {4, 5}, is a two-element set, not an interval.

Q14 (8 pts) — $|(B \times A) \setminus (A \times B)|$. Answer: D (26) — Phase 2 §7

A = {a, b, c, d, e} (5 elements), B = {d, e, f, g, h, i} (6 elements), A ∩ B = {d, e}.

- $|B \times A| = 6 \times 5 = 30$.
- Pairs in both products need both parts from A ∩ B: $2 \times 2 = 4$.
- $30 - 4 =$ **26**.

Q15 (5 pts) — Error in the truth table. Answer: C (Row 3) — Phase 4 §8

Row 3 (p = F, q = T): ¬q = F; p ⇒ ¬q = F ⇒ F = T; ¬(p ⇒ ¬q) = F; F ∨ p = F ∨ F = **F**. The table shows T in that column. Every other row checks out.

Mock questions 17 to 23

Q17 (7 pts) — Logical consequence. Answer: B — Phase 4 §6 and §8

B is the chain rule: from p ⇒ q and q ⇒ r, p ⇒ r always follows (the full 8-row table is all T in Phase 4).

The others fail:

- (A) ¬q makes p ∧ q false.
- (C) ¬p makes p ∧ q false.
- (D) q ∧ ¬p makes p false.

Q18 (6 pts) — When is $\neg\neg((p \Rightarrow q) \vee (q \Rightarrow p))$ false? Answer: A (never) — Phase 4 §8

¬¬ cancels. One of p ⇒ q and q ⇒ p is always true (if q = T the first is true; if q = F the second is true). So the formula is valid: never false.

Q19 (7 pts) — Recursion, find $g(4)$. Answer: C (109) — Phase 3 §9

$g(0) = 1$, $g(1) = 2$, $g(n) = 3g(n - 1) - g(n - 2) + n^2$.

- $g(2) = 3(2) - 1 + 4 = 9$

- g(3) = 3(9) − 2 + 9 = 34
- g(4) = 3(34) − 9 + 16 = **109**

Q20 (2 pts) — Graph of h(x) = ½x − ¾. Answer: B — Phase 3 §5

It rises gently (slope ½), crosses the y-axis at −0.75 (just below the origin) and the x-axis at 1.5. Only graph B matches. A is too steep, C passes through the origin, and D falls.

Q21 (3 pts) — Inverse of $x^2 + 2$, $x \geqslant 0$. Answer: A ($\sqrt{(x - 2)}$) — Phase 3 §7

Swap: $x = y^2 + 2 \rightarrow y^2 = x - 2 \rightarrow y = \sqrt{(x - 2)}$. Check: f(3) = 11, and $\sqrt{(11 - 2)} = 3$. ✓

Q22 (3 pts) — 45 in binary. Answer: A (101101) — Phase 1 §1

45 = 32 + 8 + 4 + 1 → 101101. Shortcut: 45 is odd, so it must end in 1, which rules out D. It needs 6 digits, which rules out B.

Q23 (7 pts) — Encrypt "dog" with (4P + 1) mod 26. Answer: C (nfz) — Phase 1 §6

- d = 3 → 13 = n
- o = 14 → 57 mod 26 = 5 = f
- g = 6 → 25 = z

One-page method sheet (read this the morning of the exam)

Numbers (Phase 1)

Topic	Method
Decimal → binary	Subtract the largest powers of 2 (1, 2, 4, 8, 16, 32, 64, 128, 256); a 1 for each used, a 0 for each skipped
Binary → decimal	Label positions 1, 2, 4, 8 … from the right; add where there's a 1
Primes	Greater than 1, divisible only by 1 and itself. 1 is not prime; 2 is the only even prime
Prime factorisation	Divide by the smallest prime repeatedly; write as powers ($2^2 \times 3 \times 7$)

Topic	Method
Division theorem	$a = qb + r$, with $0 \leqslant r < \lvert b \rvert$. Full blocks = q; leftover = r; empty space in last block = $b - r$
a mod b	The remainder, always 0 to $b - 1$. Negative a: add b until it's in range
Wrap-around	(start ± move) mod size. Numbered from 1: a result of 0 means the last item
Cipher	a = 0 … z = 25. Apply the formula, then mod 26
Arithmetic sequence	$a_n = a + (n - 1)d$
Geometric sequence	$a_n = a \cdot r^{n-1}$
Fibonacci-type	Each term = sum of the previous two

Sets and intervals (Phase 2)

Topic	Method
$\lvert A \rvert$	Count top-level elements; inner sets count as 1; duplicates (including repeated { }) count once
$\in$ vs $\subseteq$	$\in$: element in set. $\subseteq$: every element of the left set is in the right set. $\varnothing \subseteq$ every set
$\subset$	Subset and not equal
$\cup \cap \setminus '$	Or / and / in first but not second / not in A (within U)
Simplify	$A \cup A' = U$, $A \cap A' = \varnothing$, $A \cap U = A$, $A \cup \varnothing = A$, $U' = \varnothing$, $\varnothing' = U$
$\lvert P(S) \rvert$	2^n
$\lvert A \times B \rvert$	$\lvert A \rvert \times \lvert B \rvert$; order of pairs matters
$\lvert (A \times B) \setminus (B \times A) \rvert$	$\lvert A \rvert\lvert B \rvert - \lvert A \cap B \rvert^2$
$\lvert A \cup B \rvert$	$\lvert A \rvert + \lvert B \rvert - \lvert A \cap B \rvert$

Topic	Method
Interval brackets	( ) excludes the end (< or >); [ ] includes it (⩽ or ⩾); ∞ always ( )
{ } vs [ ]	{1, 2} has two elements; [1, 2] has infinitely many
$\{x \in A : g(x) \in A\}$	Solve low ⩽ g(x) ⩽ high, then intersect with A

Functions (Phase 3)

Topic	Method
Is it a function from A?	Every element of A used exactly once; outputs in B
Slope	$m = (y_2 - y_1)/(x_2 - x_1)$
Line	y = mx + b; find b by substituting a point
Intercepts	y-axis: x = 0 gives (0, b). x-axis: y = 0 gives (−b/m, 0)
Parallel	Same slope
Perpendicular	$m_2 = -1/m_1$ (flip and change the sign)
Inverse	Swap x and y, solve for y; check with a test number
Intersection	Set f(x) = g(x), factorise, substitute back
Recursion	Base case first, then a table row by row up to exactly the term asked
n!	0! = 1, 1, 2, 6, 24, 120, 720
Binary search	$M = \lfloor (L + R)/2 \rfloor$; too small → L = M + 1; too big → R = M − 1
Bubble sort pass	Compare neighbours left to right; swap if the left is bigger

Logic (Phase 4)

p	q	¬p	p ∧ q	p ∨ q	p → q
T	T	F	T	T	T
T	F	F	F	T	F
F	T	T	F	T	T
F	F	T	F	F	T

- Precedence: ¬, then ∧, then ∨, then →.
- Rows = 2^n, where n = number of **different** letters.
- Valid = all T. Satisfiable = at least one T. Unsatisfiable = no T.
- C follows from KB when $(KB_1 \wedge KB_2 \wedge \ldots) \rightarrow C$ is valid. Equivalently, there is no row where all KB facts are T and C is F.
- $\neg(p \rightarrow q) \equiv p \wedge \neg q$. $p \rightarrow q \equiv \neg p \vee q \equiv \neg q \rightarrow \neg p$.

Harder practice exam (26 questions, about 60 minutes)

No notes, no calculator. Mark your answers, then check the next section. These are deliberately a step harder than the mock.

Numbers

1. 156 in binary is: (A) 10011100 (B) 10011010 (C) 11001100 (D) 1001110
2. What is $101011_2 + 1101_2$, written in decimal? (A) 54 (B) 56 (C) 58 (D) 112
3. −45 mod 8 = (A) −5 (B) 5 (C) 3 (D) −3
4. Nine servers are numbered 1 to 9 and take requests in turn, wrapping around. Request #1 went to server 7. Which server gets request #51? (A) 3 (B) 4 (C) 2 (D) 9
5. Letters are numbered a = 0 to z = 25 and encrypted with C = (3P + 1) mod 26. A ciphertext letter is "k". What was the plaintext letter? (A) c (B) d (C) e (D) j
6. A geometric sequence with a positive ratio has 2nd term 12 and 4th term 108. What is its 6th term? (A) 324 (B) 972 (C) 2916 (D) 486
7. An arithmetic sequence has 3rd term 11 and 7th term 27. What is its 20th term? (A) 79 (B) 83 (C) 76 (D) 80
8. The prime factorisation of 2310 is: (A) 2 × 3 × 5 × 7 × 11 (B) 2 × 3 × 5 × 77 (C) $2^2 \times 3 \times 5 \times 7 \times 11$ (D) 10 × 231

Sets and intervals

9. $|\{\varnothing, \{\varnothing\}, \{\varnothing, \{\varnothing\}\}, \varnothing\}|$ = (A) 2 (B) 3 (C) 4 (D) 5
10. $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5\}$. $|P(A \cap B)|$ = (A) 2 (B) 4 (C) 8 (D) 16
11. $A = \{x \in \mathbb{Z} : -2 \leqslant x < 3\}$ and $B = \{x \in \mathbb{N} : x^2 < 10\}$, with $\mathbb{N} = \{1, 2, 3, \ldots\}$. Find $A \setminus B$ and $|A \times B|$.
12. $A = \{1, 2, \ldots, 6\}$ and $B = \{4, 5, \ldots, 9\}$. $|(A \times B) \setminus (B \times A)|$ = (A) 36 (B) 33 (C) 27 (D) 30
13. Which statement is true? (A) $(2, 5) \subseteq [3, 5]$ (B) $[2, 3] \subset (1, 4)$ (C) $\{2, 3, 4\} \subseteq (2, 4]$ (D) $[1, 4] \cap (4, 6) = \{4\}$
14. $A = [1, 16]$. Which interval is $\{x \in A : \sqrt{x} \in A \text{ and } x/2 \in A\}$? (A) $[1, 16]$ (B) $[2, 16]$ (C) $[2, 32]$ (D) $[1, 4]$
15. Simplify $((A \cup \varnothing) \cap (A \cup A'))' \cup (U \setminus A')$. (A) A (B) A' (C) U (D) $\varnothing$

Functions and algorithms

16. $A = \{1, 2, 3\}$, $B = \{a, b, c\}$. Which function from A to B has an inverse? (A) {(1, a), (2, a), (3, b)} (B) {(1, b), (2, c), (3, a)} (C) {(1, a), (2, b)} (D) {(1, c), (1, a), (2, b), (3, c)}
17. Find the line through $(-1, 4)$ that is perpendicular to the line $2x + 3y = 6$. (Hint: first rewrite it as $y = \ldots$)
18. The inverse of $f(x) = (5 - 2x)/3$ is: (A) $(3x - 5)/2$ (B) $(5 - 3x)/2$ (C) $(5 + 3x)/2$ (D) $3/(5 - 2x)$
19. Where do $y = x^2 - 3x + 2$ and $y = x - 1$ intersect?
20. $u(0) = 1$, $u(1) = 3$; $u(n) = u(n - 1) \times u(n - 2) - n$ for $n \geqslant 2$. What is $u(5)$? (A) −5 (B) −4 (C) 0 (D) 5
21. $f(0) = 0$; $f(n + 1) = f(n) + (n + 1) \times (n + 1)!$. What is $f(4)$? (A) 23 (B) 119 (C) 120 (D) 96
22. How many comparisons does binary search need to find 3 in 3, 7, 11, 23, 27, 31, 37, 43, 47? (A) 1 (B) 2 (C) 3 (D) 4

Logic

23. How many times does TRUE appear in the final column of $(p \to q) \wedge (q \to r)$? (A) 2 (B) 3 (C) 4 (D) 5
24. $KB = \{p \vee q, p \to r, q \to r\}$. Is r a logical consequence?
25. $\neg(p \to \neg q)$ is equivalent to: (A) $p \wedge q$ (B) $\neg p \vee q$ (C) $p \wedge \neg q$ (D) $\neg p \wedge \neg q$
26. A student's truth table for $\neg p \vee (p \wedge q)$ is below. Which row contains a mistake?

Row	p	q	¬p	p ∧ q	¬p ∨ (p ∧ q)
1	T	T	F	T	T
2	T	F	F	F	F

Row	p	q	¬p	p ∧ q	¬p ∨ (p ∧ q)
3	F	T	T	T	T
4	F	F	T	F	T

Practice exam answers with full explanations

Numbers

1. **(A)** 156 = 128 + 16 + 8 + 4. Positions 128, 64, 32, 16, 8, 4, 2, 1 → 1, 0, 0, 1, 1, 1, 0, 0 = 10011100. (D) has only 7 digits, so it's too small.
2. **(B)** 101011_2 = 32 + 8 + 2 + 1 = 43. 1101_2 = 8 + 4 + 1 = 13. 43 + 13 = 56.
3. **(C)** 45 mod 8 = 5, and 8 − 5 = 3. Check: −45 = (−6) × 8 + 3.
4. **(A)** Request #51 is 50 steps after request #1. 7 + 50 = 57. 57 mod 9 = 3 (9 × 6 = 54). Server 3. (A result of 0 would have meant server 9.)
5. **(B)** We need 3P + 1 ≡ 10 (mod 26). Try candidates: P = 3 gives 10 = k. ✓ So the letter was d. Checking the options directly is fastest: c → 7 = h, d → 10 = k, e → 13 = n, j → 28 mod 26 = 2 = c.
6. **(B)** The 4th term is the 2nd term × r^2: 108 = 12r^2, so r^2 = 9 and r = 3. First term a = 12/3 = 4. $a_6 = 4 \times 3^5 = 4 \times 243 = 972$.
7. **(A)** From the 3rd to the 7th term is 4 steps: 27 − 11 = 16 = 4d, so d = 4. a = 11 − 2(4) = 3. $a_{20} = 3 + 19 \times 4 = 79$.
8. **(A)** 2310 ÷ 2 = 1155, ÷ 3 = 385, ÷ 5 = 77, ÷ 7 = 11. All prime. (B) and (D) contain non-primes; (C) multiplies to 4620.

Sets and intervals

9. **(B)** Top-level pieces: ∅, {∅}, {∅, {∅}}, ∅. The two ∅ are duplicates. 3 distinct elements.
10. **(B)** A ∩ B = {3, 4}, which has 2 elements. 2^2 = 4.
11. A = {−2, −1, 0, 1, 2}. B = {1, 2, 3} (1, 4 and 9 are under 10; 16 isn't). **A \ B = {−2, −1, 0}. |A × B| = 5 × 3 = 15.**
12. **(C)** |A| = |B| = 6. A ∩ B = {4, 5, 6}, 3 elements. 6 × 6 − 3^2 = 36 − 9 = 27.
13. **(B)** [2, 3] sits inside (1, 4) with room at both ends, and they're not equal. (A) fails at 2.5; (C) fails because 2 is excluded; (D) is ∅ because 4 is excluded from (4, 6).
14. **(B)** √x ∈ [1, 16] gives x ∈ [1, 256]. x/2 ∈ [1, 16] gives x ∈ [2, 32]. Intersect both with A = [1, 16]: [2, 16].

15. **(C)** $A \cup \varnothing = A$. $A \cup A' = U$. $A \cap U = A$, and its complement is A'. $U \setminus A' = A$. $A' \cup A = U$.

Functions and algorithms

16. **(B)** It needs to be a function that is one-to-one and onto. (B) sends 1, 2, 3 to b, c, a: each output used exactly once. (A) isn't one-to-one; (C) leaves 3 out; (D) gives 1 two outputs.
17. Rearrange: $3y = 6 - 2x$, so $y = -(2/3)x + 2$ and the slope is $-2/3$. Perpendicular slope: $3/2$. Then $4 = (3/2)(-1) + b$, so $b = 4 + 1.5 = 5.5$. **$y = (3/2)x + 11/2$.**
18. **(B)** $x = (5 - 2y)/3 \rightarrow 3x = 5 - 2y \rightarrow 2y = 5 - 3x \rightarrow y = (5 - 3x)/2$. Check: $f(1) = 1$ and $(5 - 3)/2 = 1$. ✓
19. $x^2 - 3x + 2 = x - 1 \rightarrow x^2 - 4x + 3 = 0 \rightarrow (x - 1)(x - 3) = 0$. $x = 1$ gives $y = 0$; $x = 3$ gives $y = 2$. **(1, 0) and (3, 2).**
20. **(A)** $u(2) = 3 \times 1 - 2 = 1$. $u(3) = 1 \times 3 - 3 = 0$. $u(4) = 0 \times 1 - 4 = -4$. $u(5) = (-4) \times 0 - 5 = -5$.
21. **(B)** $f(1) = 0 + 1 \times 1! = 1$. $f(2) = 1 + 2 \times 2! = 5$. $f(3) = 5 + 3 \times 3! = 23$. $f(4) = 23 + 4 \times 4! = 23 + 96 = 119$. (Pattern: $f(n) = (n + 1)! - 1$. (C), 120, is the trap.)
22. **(C)** L = 1, R = 9: M = 5 → 27 > 3, so R = 4. M = 2 → 7 > 3, so R = 1. M = 1 → 3. Found after 3 comparisons.

Logic

23. **(C)**

p	q	r	p → q	q → r	both
T	T	T	T	T	**T**
T	T	F	T	F	F
T	F	T	F	T	F
T	F	F	F	T	F
F	T	T	T	T	**T**
F	T	F	T	F	F
F	F	T	T	T	**T**
F	F	F	T	T	**T**

Four T values.

24. **Yes.** Fast method: if r = F, then p → r forces p = F and q → r forces q = F. But then p ∨ q = F, so the KB isn't all true. So there's no row where the whole KB is T and r is F. r follows. (This is reasoning by cases: either p or q holds, and each leads to r.)
25. **(A)** ¬(p → ¬q) ≡ p ∧ ¬(¬q) ≡ p ∧ q, using ¬(A → B) ≡ A ∧ ¬B.
26. **Row 3.** With p = F, q = T, p ∧ q is **F**, not T. The final value T happens to be correct anyway, because ¬p = T.

Scoring

- **22 or more:** excellent. You're ready for harder-than-mock questions.
- **17–21:** solid. Re-read the sections behind your mistakes.
- **Below 17:** go back over Phases 2 and 4 first. Sets and logic carry the most marks in your mock.

Good luck tomorrow. Get some sleep: a rested brain makes far fewer slips on truth tables and negative mods than a tired one.